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1.0 Project Overview and Data Preparation

This project develops an analytical framework to assess and enhance loan approval policies for a consumer lender. It aims to balance profitability and risk using historical LendingClub data to examine borrower behaviour, test alternative decision rules, and build predictive models for default forecasting. The analysis includes three main tasks: a simulated A/B test of loan review policies, cluster analysis to identify borrower segments, and a prototype deep learning model for default prediction. Together, these components link data driven evaluation to strategic goals, showing how approval policies influence credit losses and portfolio performance.

The dataset contains about 50,000 personal loans with borrower characteristics, loan details, and repayment outcomes. To ensure realistic assessment, only origination time variables were retained. Identifiers, text fields, post-outcome features, and redundant columns were removed, resulting in 49,969 loans with 28 variables and no missing data. Extreme outliers in loan amount, income, and DTI were trimmed using practical underwriting limits to maintain credible ranges. The next section applies the processed dataset to compare two loan review policies through a simulated A/B test.

2.0 A/B Testing of Loan Review Policies

2.1 Policy Definition

The two policies were designed using grade and debt-to-income ratio (DTI), both ranking in the top five variables by information gain (IG). The selection of grade as the primary variable is supported by Serrano-Cinca *et al.* (2015) and Emekter *et al.* (2015), who identify it as the strongest predictor of default in LendingClub data. The inclusion of DTI is further supported by Polena and Regner (2018), who confirm it as a significant determinant of default across all loan risk classes. Although interest rate was strongly correlated with default, it was excluded because it largely duplicates the information captured by grade.

Policy A (Control)

Approve if grade $\in$ {A, B, C, D} and DTI $\leq$ 25%.

Reject otherwise.

Policy B (Treatment)

Approve if grade ∈ {A, B, C} and DTI ≤ 20%.

Reject otherwise.

Grades A–C represent stronger borrowers, while grade D includes moderate-risk applicants (Figure 1). In the dataset, a 20 percent DTI is close to the median and 25 percent to the upper quartile (Figure 2). Policy A thus aims for higher approval volume by allowing slightly weaker, more leveraged borrowers, whereas Policy B focuses on financial strength and lower risk exposure.

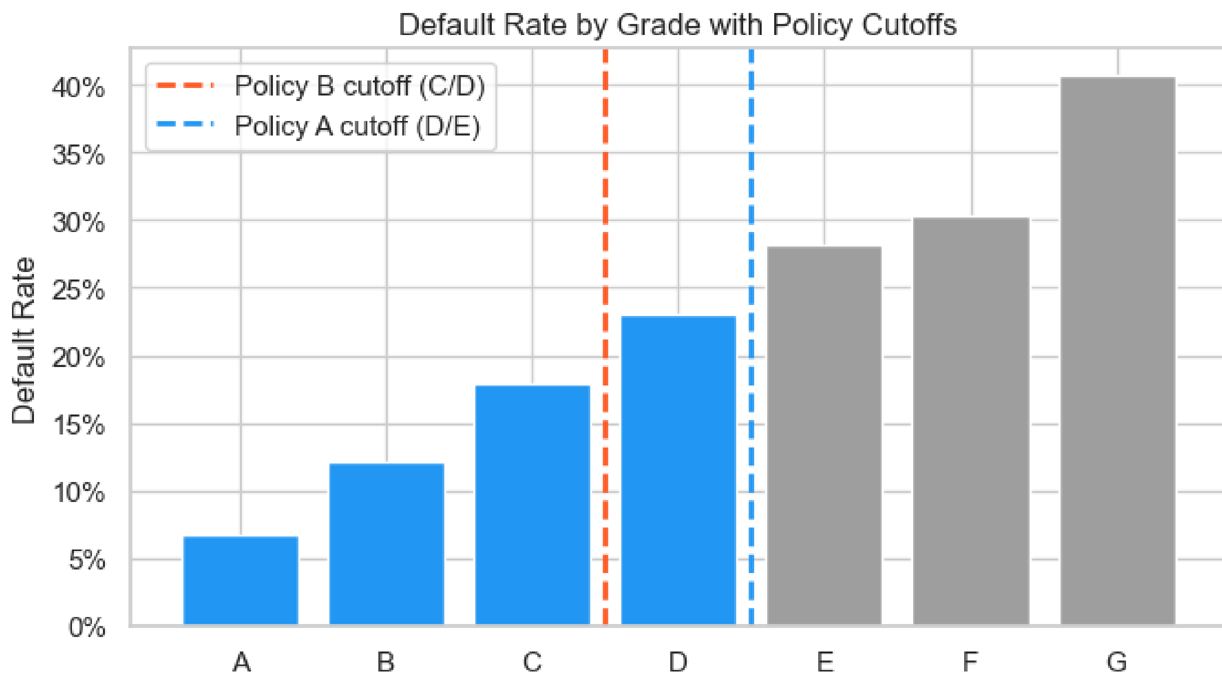

Figure 1: Default Rate by Grade with Policy Cutoffs

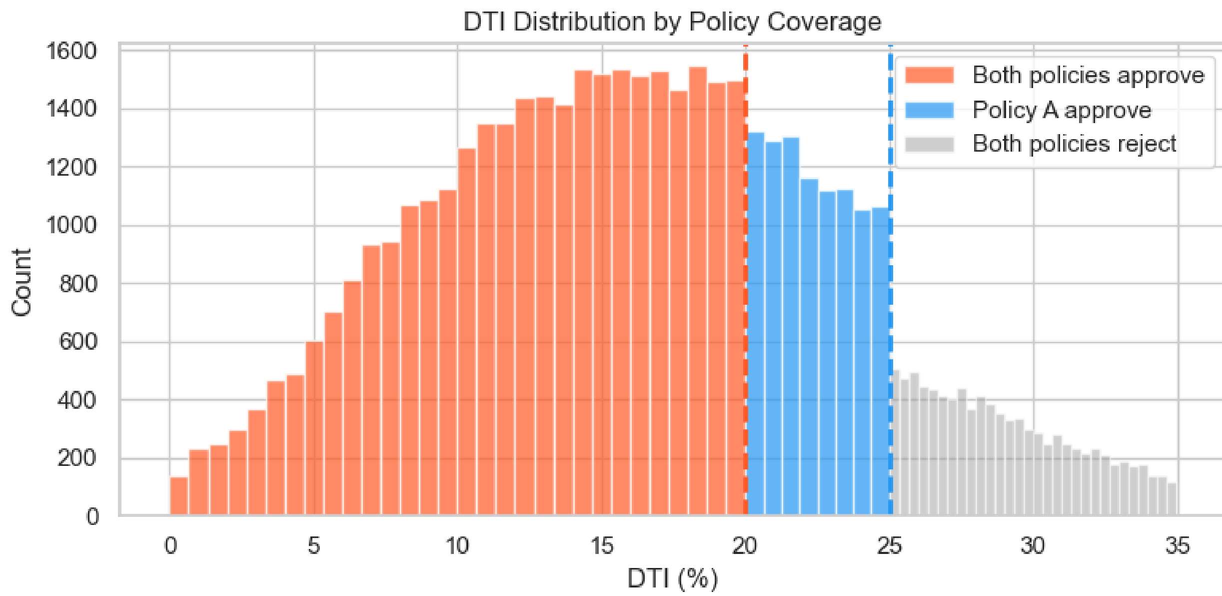

Figure 2: DTI Distribution by Policy Coverage

2.2 Evaluation Metric and Hypotheses

The primary metric is the default rate among approved loans:

$$Default\ Rate\ = \frac{bad\ loans\ approved}{total\ loans\ approved}$$

This directly measures portfolio credit quality where lower values indicate better credit performance and fewer losses, aligning with the company's objective to minimise default losses.

Hypotheses

- H_0: $Policy\ A\ =\ Policy\ B$ — No difference in default rates.
- H_1: $Policy\ A\ >\ Policy\ B$ — Policy A's default rate is higher, indicating Policy B performs better.

Significance level: α = 0.05, one-tailed upper two-proportion z-test.

2.3 Policy Comparison

Using 49,969 historical loans, the simulated results were:

Metric	Policy A (Control)	Policy B (Treatment)
Grades Approved	A-D	A-C
DTI Limit	<=25%	<=20%
Loans Approved	37,438	24,905
Approval Rate	74.90%	49.80%
Default Rate	13.51%	11.33%

Table 1: Results for Policy A and Policy B

The z-test yields **z ≈ 8.0**, **p < 0.001,** strongly rejecting H_0. Policy B reduces the portfolio default rate by 2.18 percentage points relative to Policy A. The **Cohen's h ≈ 0.066** suggests a small effect statistically, which is expected given both policies already exclude the highest-risk grades (E, F, G). However, the practical cost is 12,533 for fewer approved loans, a trade-off between capital preservation and foregone interest revenue.

Business Implications

Policy B strengthens portfolio quality with only a moderate decline in approval rate, making it appropriate for a risk-controlled or uncertain market environment. Policy A may still be chosen when the business's goal is to drive short-term growth, but it would require tighter monitoring and higher capital buffers to offset higher loss levels.

2.4 Limitations

The evaluation uses historical observational data rather than a true experiment, so results reflect correlations and not proven causation. Borrower behaviour under the new policies might differ, and macroeconomic conditions at the time could have influenced defaults in ways this simulation cannot isolate. The dataset also captures only loans that were funded, leaving applicants who might have been approved or rejected differently under alternative policies. Additionally, default rate alone does not capture profitability; a full evaluation requires weighing reduced default losses against foregone interest income from the 12,533 rejected loans. To confirm causality before large-scale adoption, the next step should be a controlled online A/B trial assigning new applications randomly between the two policies and tracking real default outcomes over time.

3.0 Borrower Segmentation via Cluster Analysis

While Section 2 compared policy performance at the aggregate level, borrowers vary considerably in their financial profiles and risk characteristics. Cluster analysis is therefore applied to group borrowers with similar characteristics, identifying meaningful segments and examining whether policy performance differs across them.

3.1 Feature Selection and Methodology

Clustering was based on borrower information known at loan approval: debt-to-income ratio (DTI), annual income, revolving utilization, delinquencies in the past two years, loan amount, and loan term.

"These variables were chosen because they collectively capture repayment ability, financial pressure, past credit behaviour, and loan exposure, providing a comprehensive picture of borrower risk at origination."

The default outcome and LendingClub grade were excluded to keep the analysis unbiased. Grade already summarises risk, so including it would limit the ability to identify independent borrower groups.

There were no extreme correlations between variables, so all were retained (see Figure 3).

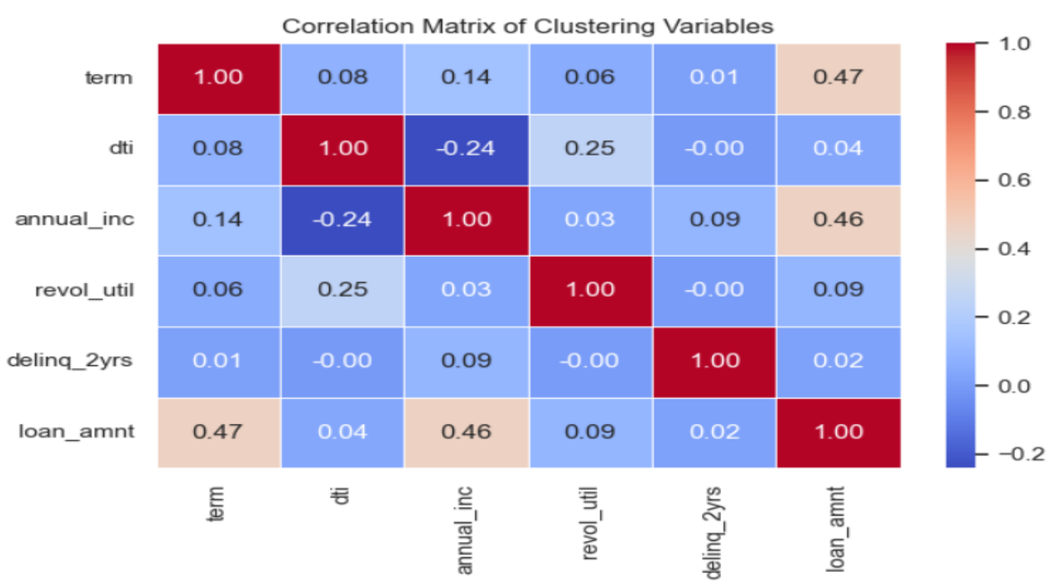

Figure 3: Correlation Matrix of Clustering Variables

K-means was selected because it works well with numerical data, is efficient for large datasets, and produces clear and easy-to-interpret cluster averages. Since K-means is distance-based, continuous variables were standardized to ensure they were on the same scale. The binary term variable remained unchanged.

The number of clusters was chosen using the Elbow method and Silhouette score (see Figures 4 and 5). Both suggested that three clusters provide clear separation without overcomplicating the model. A four-cluster solution was also tested, but it produced one relatively smaller cluster, so three clusters were selected.

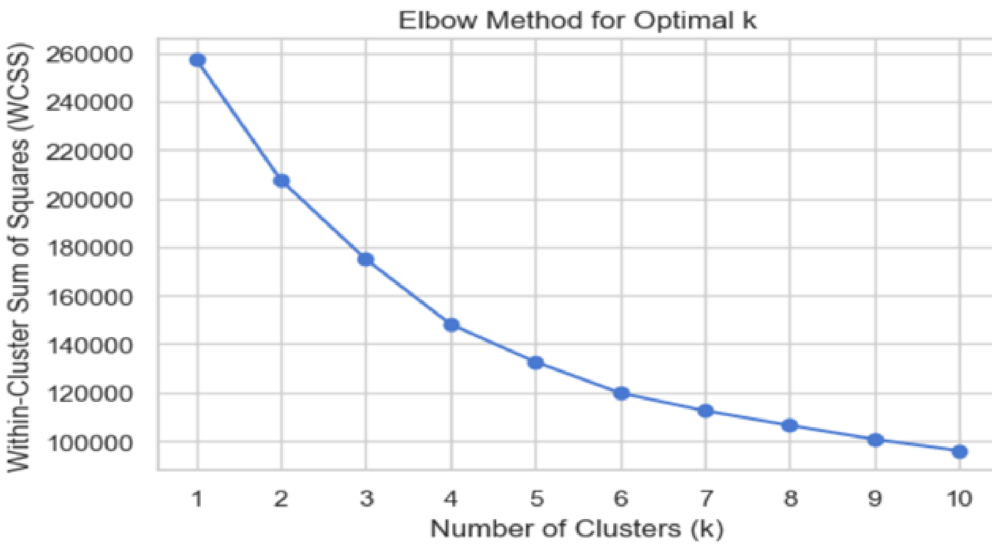

Figure 4: Elbow Method for Optimal k

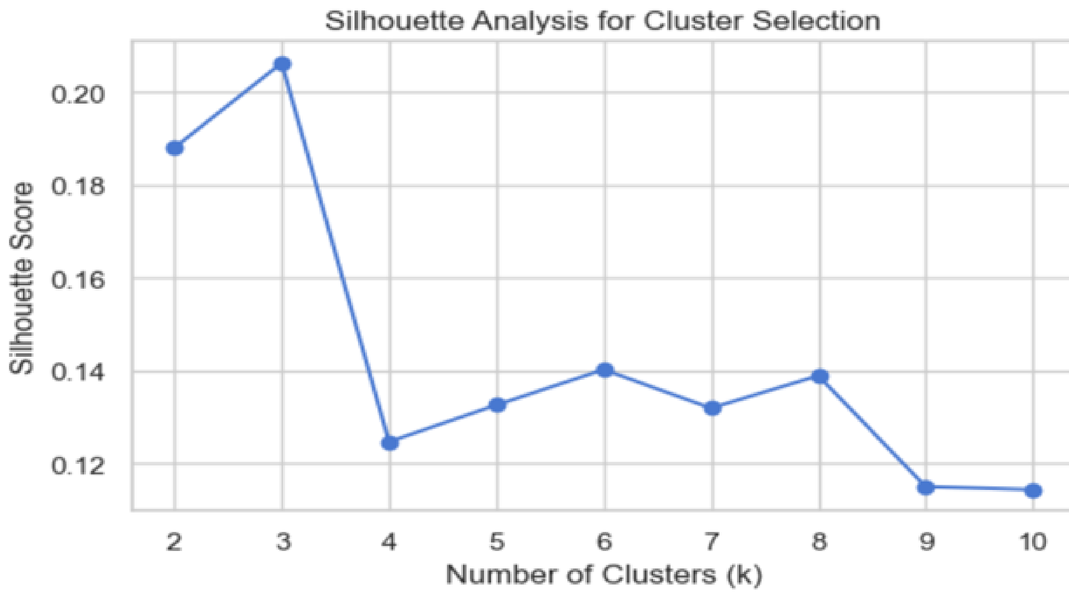

Figure 5: Silhouette Method for Optimal k

3.2 Cluster Interpretation

Three borrower segments were identified:

Cluster 0 - High Income, High Loan Size (Moderate Risk)

These borrowers earn the most (around £113k) but also take larger loans and longer terms. Their default rate is 15.4%. Although they have a strong income, their higher exposure increases risk.

Cluster 1 - Financially Stretched Borrowers (High Risk)

This is the largest group. Borrowers have the highest DTI, higher utilisation, and lower income. Their default rate is 18.3%, the highest among all clusters.

Cluster 2 - Conservative Borrowers (Low Risk)

These borrowers have the lowest DTI, smaller loans, and shorter terms. Their default rate is 12.3%, the lowest in the portfolio.

The difference between the highest- and lowest-risk clusters is more than six percentage points, which is financially meaningful.

3.3 Policy Performance by Cluster and Business Implications

Policy A and Policy B were then tested within each cluster (see Figure 6). In all three segments, Policy B yields lower default rates than Policy A, and the differences are statistically significant.

Figure 6: Default Rates by Policy and Cluster

The largest improvement under Policy B occurs in Cluster 1, confirming that stricter criteria have the greatest impact on financially stretched borrowers. Since risk is unevenly distributed across segments, applying a single approval threshold to all borrowers is suboptimal. A segmented underwriting approach, with stricter criteria applied to Clusters 0 and 1 and looser criteria applied to Cluster 2, would better balance risk control and profitability by preserving low-risk approvals while reducing losses where they are most concentrated.

4.0 Deep Learning for Default Prediction

4.1 Prediction Task and Data Split

While rule-based policies help reduce default risk and reveal variation across borrower segments, these fixed thresholds cannot capture the more complex relationships between individual borrower characteristics. To address this, a deep learning model is implemented to predict each applicant's probability of default, framed as a binary classification task, enabling more detailed and flexible loan approval decisions.

Nine origination-level features were selected based on mutual information scores: loan amount, term, interest rate, DTI, annual income, revolving utilisation, inquiries in the last six months, delinquencies in the past two years, and employment length. Grade was deliberately excluded so the model learns from underlying borrower characteristics rather than LendingClub's internal rating.

The dataset is partitioned into training (70%), validation (15%), and test (15%) sets using stratified sampling to ensure that the original class distribution (approximately 14.7% defaults) is preserved across all subsets.

4.2 Model Design and Training

Three architectures of increasing complexity were trained and compared, with full details in Table 2. All models used the Adam optimiser (lr = 0.001) and early stopping on validation loss.

Model	Architecture	Loss Function	Regularisation
Model 1	Input → 16 → 1	BCEWithLogitsLoss + pos_weight	None
Model 2	Input → 32 → 16 → 1	BCEWithLogitsLoss + pos_weight	None
Model 3	Input → 64 → 32 → 16 → 1	BCEWithLogitsLoss + pos_weight	BatchNorm + Dropout(0.3)

Table 2: Model Architecture

To address class imbalance, sampling techniques including undersampling (RandomUnderSampler) and oversampling (SMOTE) were explored. However, both approaches resulted in poorer model performance, likely due to distortion of the original data distribution. The final approach uses a positive class weight within BCEWithLogitsLoss, which increases the penalty for misclassifying default cases, thereby improving the model's ability to detect high-risk borrowers while preserving the full dataset.

Overall, this design balances model complexity and regularisation while addressing class imbalance in a way that maintains the integrity of the underlying data.

4.3 Model Evaluation

In terms of model performance, all three models achieve near-identical performance, demonstrating that greater depth does not automatically translate to better predictive power on this dataset which is a common outcome when the signal in origination-level features is limited. Model 3 is nonetheless selected as the preferred model based on its highest recall (0.667), which is the most important criterion in a lending context were failing to identify a defaulter carries greater financial cost than a false rejection.

Model	Architecture	Test AUC	Precision	Recall	F1
Model 1	1 hidden layer	0.668	0.235	0.646	0.345
Model 2	2 hidden layers	0.666	0.236	0.655	0.347
Model 3 ✓	**Deep + regularisation**	**0.669**	**0.234**	**0.667**	**0.347**

Table 3: Model Performance on Test Set

All three models were evaluated within each borrower segment. Table 4 places their AUC scores alongside the actual default rates and Policy A/B outcomes from earlier sections, enabling a direct cross-method comparison.

Mod	Actual DR	Policy A DR	Policy B DR	M1 AUC	M2 AUC	M3 AUC ✓
0 – High Income	15.4%	14.8%	12.6%	0.685	0.692	**0.687**
1 – Financially Stretched	18.3%	17.9%	14.8%	0.641	0.644	**0.643**
2 – Conservative	12.3%	11.9%	9.8%	0.662	0.667	**0.666**
Overall Portfolio	**15.1%**	**13.51%**	**11.33%**	**0.668**	**0.666**	**0.669**

Table 4: Default Rate and Model AUC by Cluster

Cluster-level AUC values are broadly consistent across all three models, indicating that increased architectural complexity does not improve predictive performance. Model 3 performs best in Cluster 0 (High Income; AUC = 0.687) and weakest in Cluster 1 (Financially Stretched; AUC = 0.643), which also has the highest observed default rate (18.3%) and the largest reduction under Policy B. This suggests that the model finds it more difficult to distinguish between defaulters and non-defaulters in higher-risk segments.

The weaker performance in Cluster 1 likely reflects more complex, non-linear interactions between factors such as high debt-to-income ratios, lower income, and elevated credit utilisation. This indicates potential benefits from additional feature engineering or segment-specific models. In contrast, Cluster 2 (Conservative) exhibits lower default rates and moderate AUC, suggesting that lower-risk borrowers are both easier to price and to predict.

Table 5 compares actual and predicted default rates across borrower clusters for Policy A-approved loans, along with the false negative rate (FNR).

Model	Cluster	Label	Actual DR	Predicted DR	FNR
Model 1	0	Affluent	11.9%	24.5%	6.9%
Model 1	1	High-DTI	16.5%	50.7%	5.4%
Model 1	2	Low-Risk	11.9%	25.2%	6.9%
Model 2	0	Affluent	11.9%	25.8%	6.4%
Model 2	1	High-DTI	16.5%	50.2%	5.4%
Model 2	2	Low-Risk	11.9%	25.9%	6.6%
Model 3	0	Affluent	11.9%	27.2%	6.3%
Model 3	1	High-DTI	16.5%	52.4%	5.1%
Model 3	2	Low-Risk	11.9%	25.9%	6.6%

Table 5: Actual vs Predicted Default Rate and FNR by Cluster

Cluster 1 (high DTI; lower income) is the riskiest group, with the highest actual default rate (16.5%). Predicted default rates are much higher than actual rates in all clusters, reflecting the use of class weighting and a 0.5 threshold, which biases the model toward identifying potential defaulters.

FNR is lowest in Cluster 1 (5.1% for Model 3), meaning high-risk borrowers are least likely to be missed. This is important because missing these borrowers would lead to higher financial losses. However, the model is conservative and rejects many borrowers who would not default, which may reduce profits. Model 3 has the lowest FNR in all clusters, supporting its selection, although adjusting the threshold could improve the balance between risk and profitability.

5.0 Insight and Implementation

5.1 Integration

This project evaluates loan approval decisions through three complementary methods, each operating at a different level of analysis. Table 6 summarizes the key output and implication of each stage.

Method	Key Output	Implication
A/B Testing	Default rate falls from 13.51% to 11.33%	Stricter criteria improve portfolio quality but reject 12,533 loans
Cluster Analysis	Default rate ranges from 12.3% to 18.3% across segments	A uniform policy is suboptimal across different borrower types
Deep Learning	Three models of increasing complexity all score between AUC 0.666 and 0.669	Better predictions require better data at application, not a more complex model

Table 6: Summary of Methods, Key Outputs and Implications

Each method operates at a different level of analysis and builds on the previous. A/B testing establishes that stricter approval criteria reduce the portfolio default rate at the aggregate level. Cluster analysis reveals this improvement is unevenly distributed across borrower segments, with the highest-risk group benefiting most. Deep learning then addresses the limitation of fixed thresholds by scoring each applicant individually. Together, the three methods form a progressive pipeline from aggregate policy evaluation to individual risk prediction, providing the analytical foundation for a tiered approval policy that is both statistically validated and responsive to borrower heterogeneity.

5.2 Insights and Next Steps

The central insight from this analysis is that no single threshold works equally well across all borrower types. Stricter criteria reduce defaults but at the cost of rejecting viable loans, and the benefit is not evenly distributed across segments. Going forward, segment specific thresholds should be applied, with looser criteria for Cluster 2 to recover low risk rejected loans while maintaining stricter standards for Clusters 0 and 1. To further improve predictive accuracy, a dedicated model trained specifically on Cluster 1 data is needed, as no architecture tested exceeded an AUC of 0.643 for this segment. Both changes should be evaluated through a live A/B trial on incoming applications, tracking approval rates, default rates, and business performance against the current policy. If results show meaningful improvement, the structured policy and dedicated Cluster 1 model can be gradually rolled out across the full portfolio, with ongoing monitoring to ensure performance holds over time.

6.0 References

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